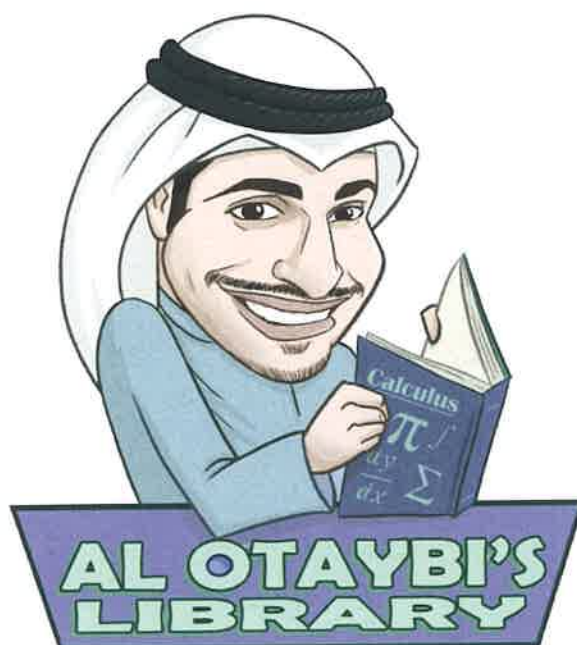


Kuwait University

2021/2022

Fall course



Calculus A

Second

Sections : 4.3 – 4.5 (Part a)

4.3 How Derivatives Affect the Shape of a Graph.

4.5 Summary of Curve Sketching.

(15)



Sketch



$$f(x)$$

Domain

Horizontal Asymptotes

Vertical Asymptotes

$$f'(x)$$

Interval of increasing and decreasing

Local Extrema

(Local maximum and Local minimum)

$$f''(x)$$

Interval of concavity

(Concave up – Concave down)

Inflection points

Key points

Intersection points with horizontal asymptote

Random points



Remark



intersection with

x-axis

y-axis

أشيل كل y وأحط مكانها صفر

أشيل كل x وأحط مكانها صفر

Example

Find the x and y intersection of

$$f(x) = \frac{x^2 - 4}{x + 1}$$

x-intercept

y-intercept

$$0 = \frac{x^2 - 4}{x + 1}$$

* $(x+1)$

$$0 = x^2 - 4$$

$$x^2 = 4$$

$$x = -2 \quad x = 2$$

$$\Rightarrow (-2, 0) \quad (2, 0)$$

$$y = \frac{0 - 4}{0 + 1}$$

$$y = \frac{-4}{1}$$

$$y = -4$$

$$\Rightarrow (0, -4)$$

Symmetry

أشيل كل x وأحط مكانها $-x$

if

$$f(-x) = -f(x)$$

Symmetry about **Origin**
(**Odd function**)

$$f(-x) = f(x)$$

Symmetry about **y-axis**
(**even function**)

Examples

Discuss the symmetry of

$$f(x) = \frac{x}{3 + x^4}$$

$$\begin{aligned} f(-x) &= \frac{(-x)}{3 + (-x)^4} \\ &= \frac{-x}{3 + x^4} \\ &= -f(x) \end{aligned}$$

$\therefore f(-x) = -f(x)$
 $\therefore f(x)$ is symmetry about origin
(odd function)

Discuss the symmetry of

$$f(x) = \frac{x^2 + 5}{2 + x^4}$$

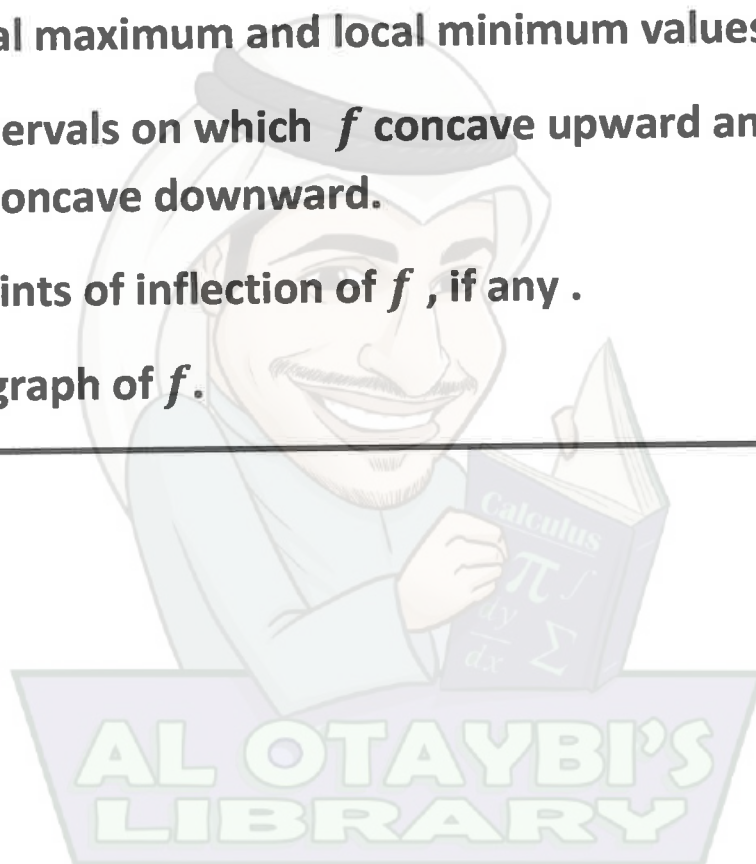
$$\begin{aligned} f(-x) &= \frac{(-x)^2 + 5}{2 + (-x)^4} \\ &= \frac{x^2 + 5}{2 + x^4} \\ &= f(x) \end{aligned}$$

$\therefore f(-x) = f(x)$
 $\therefore f(x)$ is symmetry about y-axis
(Even function)

Given that

$$f(x) = \frac{8}{x^2 - 4}, \quad f'(x) = \frac{-16x}{(x^2 - 4)^2}, \quad f''(x) = \frac{16(3x^2 + 4)}{(x^2 - 4)^3}$$

- (a) Find the vertical and horizontal asymptotes for the graph of f , if any.
- (b) Find the intervals on which f is increasing and the intervals in which f is decreasing.
- (c) Find the local maximum and local minimum values of f , if any.
- (d) Find the intervals on which f concave upward and the intervals on which f is concave downward.
- (e) Find the points of inflection of f , if any.
- (f) Sketch the graph of f .



Domain of $f(x)$: $x^2 - 4 = 0$
 $(x-2)(x+2) = 0$
 $x = 2, x = -2$
 $\Rightarrow D_f = \mathbb{R} \setminus \{\pm 2\}$

Horizontal Asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{8}{x^2 - 4} = \frac{8}{\infty} = 0 \Rightarrow y = 0 \text{ is H.A.}$$

Vertical Asymptote

لتسهيل ايجاد اشارة
نعمل تحليل المقام

$x = 2$

$$\lim_{x \rightarrow 2^+} \frac{8}{x^2 - 4} = \lim_{x \rightarrow 2^+} \frac{8}{(x-2)(x+2)} = \frac{8}{0^+} = +\infty$$

$$\lim_{x \rightarrow 2^-} \frac{8}{x^2 - 4} = \lim_{x \rightarrow 2^-} \frac{8}{(x-2)(x+2)} = \frac{8}{0^-} = -\infty$$

$\Rightarrow \therefore x = 2 \text{ is V.A.}$

$x = -2$

$$\lim_{x \rightarrow -2^+} \frac{8}{x^2 - 4} = \lim_{x \rightarrow -2^+} \frac{8}{(x-2)(x+2)} = \frac{8}{0^-} = -\infty$$

$$\lim_{x \rightarrow -2^-} \frac{8}{x^2 - 4} = \lim_{x \rightarrow -2^-} \frac{8}{(x-2)(x+2)} = \frac{8}{0^+} = +\infty$$

$\Rightarrow \therefore x = -2 \text{ is V.A.}$

$$f'(x) = \frac{-16x}{(x^2-4)^2}$$

أعداد السطوح

أعداد المقام

$$= 0$$

$$-16x = 0$$

$$\div (-16)$$

$$x = 0$$

$\in \text{Domain}$

DNE

$$x^2 - 4 = 0$$

$$(x-2)(x+2) = 0$$

$$x = 2 \quad x = -2$$

$\notin \text{Domain}$

أخذ الأعداد التي تليها

$v^+ A$



لمعرفة الإشارة
أكون بـ رقم
يسهم

$f'(x)$	$(-\infty, -2)$	$(-2, 0)$	$(0, 2)$	$(2, \infty)$
$\frac{-16x}{(x^2-4)^2}$	+	+	-	-
	+	+	+	+

+

+

-

-

أدور
OR
L.max
يعرف التماثل
للحال

- $\therefore f(x)$ is increasing on $(-\infty, -2)$ and $(-2, 0)$
- $\therefore f(x)$ is decreasing on $(0, 2)$ and $(2, \infty)$

at $x = 0 \rightarrow$ there is Local maximum

$$y = \frac{8}{0-4} = \frac{8}{-4} = -2$$

$\Rightarrow \therefore (0, -2)$ Local maximum.

$\therefore$ No Local minimum.

أنا عيب
ويطلع
أكونها
بالأداة
الأصلية

$$f''(x) = \frac{16(3x^2 + 4)}{(x^2 - 4)^3}$$

$= 0$

DNE

$$16(3x^2 + 4) = 0$$

$$\div 16$$

$$3x^2 + 4 = 0$$

$$3x^2 = -4$$

None

$$x^2 - 4 = 0$$

$$(x-2)(x+2) = 0$$

$$x = 2 \quad x = -2$$

$\notin$ Domain

أخط الأعداد إلى طالع
V.A



$f''(x)$	$(-\infty, -2)$	$(-2, 2)$	$(2, \infty)$
$\frac{16(3x^2+4)}{(x^2-4)^3}$	+	-	+
	+	-	+
	U	∩	U

أنوار
∪ ∩ OR ∪ ∩
رأى تكون كسب
Inflection point
ليس للدوميني
المعال

∴ $f(x)$ is concave upward on $(-\infty, -2)$ and $(2, \infty)$

∴ $f(x)$ is concave downward on $(-2, 2)$

⇒ ∴ No inflection point.

أساوي
بالدالة الأصلية

$$0 = \frac{8}{x^2 - 4}$$

$$0 = 8$$

None

Key points

HA

أخذ نقاط بين
نفس أراعي عدم التكرار

$$\begin{array}{c|c|c} -3 & 1 & 3 \\ \hline -2 & & 2 \end{array}$$

at $x = -3$

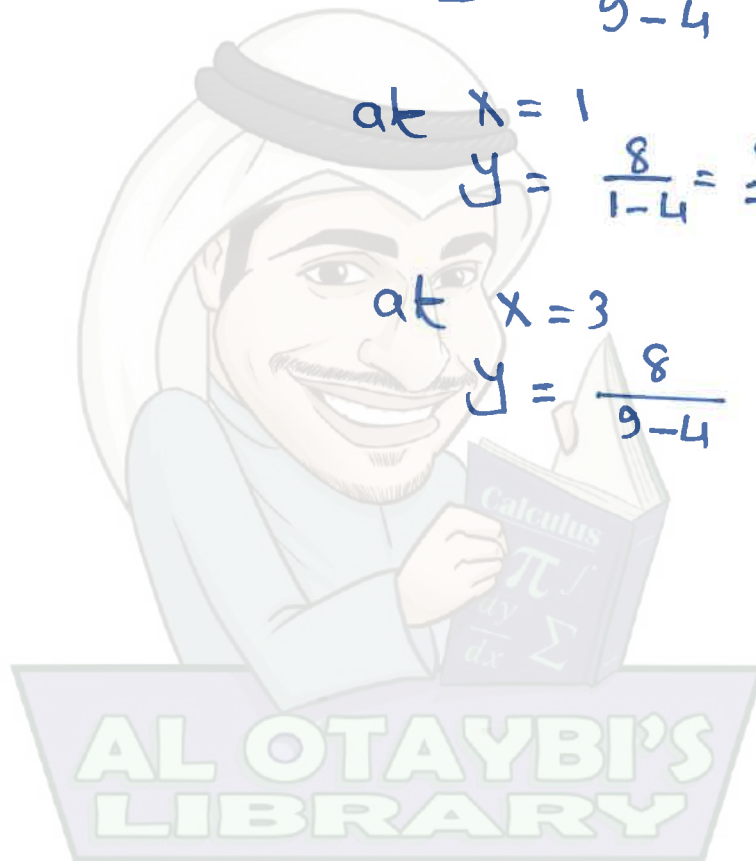
$$y = \frac{8}{9-4} = \frac{8}{5} \Rightarrow \left(-3, \frac{8}{5}\right)$$

at $x = 1$

$$y = \frac{8}{1-4} = \frac{8}{-3} = -\frac{8}{3} \Rightarrow \left(1, -\frac{8}{3}\right)$$

at $x = 3$

$$y = \frac{8}{9-4} = \frac{8}{5} \Rightarrow \left(3, \frac{8}{5}\right)$$



$$D_f = \mathbb{R} \setminus \{\pm 2\}$$

$$H.A \rightarrow y=0$$

V.A

$$x=2 \begin{cases} \oplus & +\infty \\ \ominus & -\infty \end{cases}$$

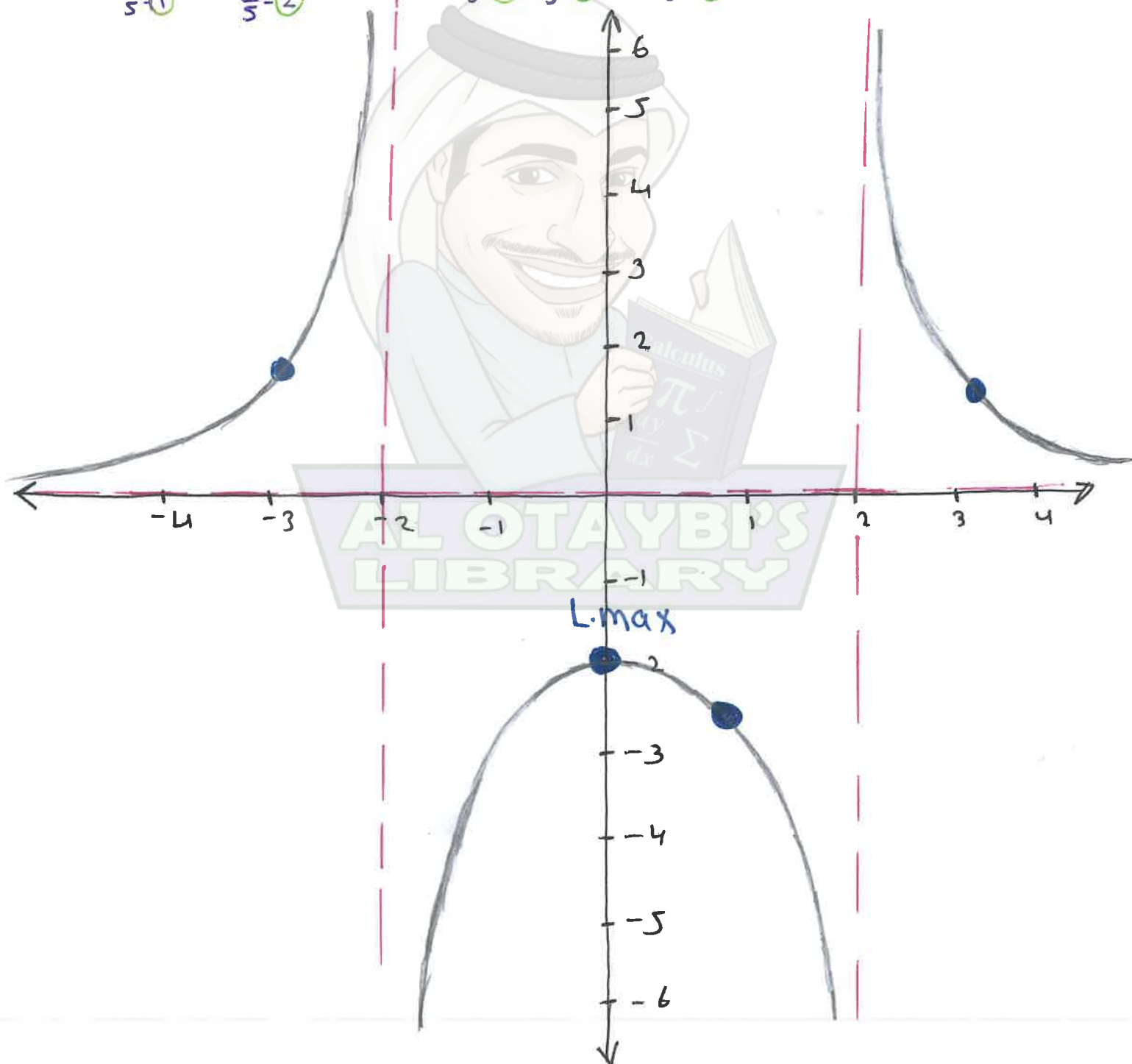
$$x=-2 \begin{cases} \oplus & -\infty \\ \ominus & +\infty \end{cases}$$

$$(0, -2) \text{ L.max.}$$

$$\cup (-\infty, -2) \quad (2, \infty) \quad \cap (-2, 2)$$

$$\left(-3, \frac{8}{3}\right) \quad \left(1, -\frac{8}{3}\right) \quad \left(3, \frac{8}{3}\right) \text{ Key points}$$

$$\frac{x}{3}=1 \quad \frac{x}{3}=2 \quad \frac{-x}{3}=-2 \quad \frac{-x}{3}=-3 \quad \frac{x}{3}=1 \quad \frac{x}{3}=2$$



Given that $f(x) = \frac{1}{x - x^2}$

(a) Find the vertical and horizontal asymptotes for the graph of f , if any.

(b) Show that $f'(x) = \frac{2x - 1}{(x - x^2)^2}$

(c) Find the intervals on which f is increasing and the intervals in which f is decreasing.

(d) Find the local maximum and local minimum values of f , if any .

Given that $f''(x) = \frac{2(3x^2 - 3x + 1)}{(x - x^2)^3}$

(e) Find the intervals on which f concave upward and the intervals on which f is concave downward.

(e) Find the points of inflection of f , if any .

(f) Sketch the graph of f .

**AL OTAYBI'S
LIBRARY**

$$f(x) = \frac{1}{x - x^2}$$

Domain of $f(x)$: $x - x^2 = 0$
 $x(1-x) = 0$
 $x = 0 \quad x = 1$

$\Rightarrow \therefore D_f = \mathbb{R} \setminus \{0, 1\}$

H.A

$$\lim_{x \rightarrow \pm\infty} \frac{1}{x - x^2} = \frac{1}{-\infty} = 0 \rightarrow \therefore y=0 \text{ is HA}$$

V.A

$x=0$
 $\lim_{x \rightarrow 0^+} \frac{1}{x(1-x)} = \frac{1}{0^+} = +\infty$
 $\rightarrow 0 \rightarrow +0.1 = 0.1 \rightarrow \therefore x=0 \text{ is V.A}$

$\lim_{x \rightarrow 0^-} \frac{1}{x(1-x)} = \frac{1}{0^-} = -\infty$
 $\rightarrow 0 \rightarrow -0.1 = -0.1$

$x=1$
 $\lim_{x \rightarrow 1^+} \frac{1}{x(1-x)} = \frac{1}{0^-} = -\infty$
 $\rightarrow 1 \rightarrow 1+0.1 = 1.1 \rightarrow \therefore x=1 \text{ is V.A}$

$\lim_{x \rightarrow 1^-} \frac{1}{x(1-x)} = \frac{1}{0^+} = +\infty$
 $\rightarrow 1 \rightarrow 1-0.1 = 0.9$

$$f(x) = \frac{1}{x - x^2}$$

$$f'(x) = \frac{0 - (1 - 2x)}{(x - x^2)^2} = \frac{2x - 1}{(x - x^2)^2}$$

$$f'(x) = \frac{2x - 1}{(x - x^2)^2}$$

= 0

$$2x - 1 = 0$$

$$2x = 1$$

$$x = \frac{1}{2}$$

∈ Domain

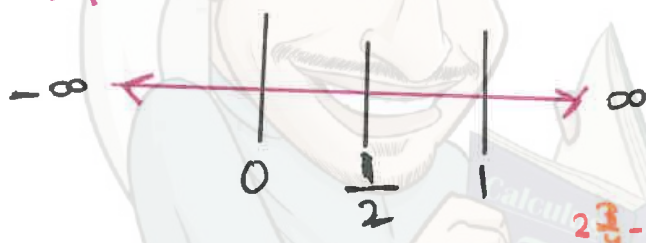
DNE

$$x - x^2 = 0$$

$$x(1 - x) = 0$$

$$x = 0 \quad x = 1$$

∉ Domain



$f'(x)$	$(-\infty, 0)$	$(0, \frac{1}{2})$	$(\frac{1}{2}, 1)$	$(1, \infty)$
$\frac{2x-1}{(x-x^2)^2}$	-	-	+	+
	+	+	+	+

∴ $f(x)$ is increasing on $(\frac{1}{2}, 1)$ $(1, \infty)$

∴ $f(x)$ is decreasing on $(-\infty, 0)$ $(0, \frac{1}{2})$

at $x = \frac{1}{2} \rightarrow$ there is Local minimum

$$y = \frac{1}{\frac{1}{2} - \frac{1}{4}} = \frac{1}{\frac{2}{4} - \frac{1}{4}} = \frac{1}{\frac{1}{4}} = 4 \Rightarrow \left(\frac{1}{2}, 4\right) \text{ L.min.}$$

$$f''(x) = \frac{2(3x^2 - 3x + 1)}{(x - x^2)^3}$$

$= 0$

DNE

$$2(3x^2 - 3x + 1) = 0$$

$\div 2$

$$3x^2 - 3x + 1 = 0$$

$$a=3 \quad b=-3 \quad c=1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{+3 \pm \sqrt{9 - 4(3)(1)}}{2(3)}$$

$$= \frac{3 \pm \sqrt{-3}}{6}$$

None

$$x - x^2 = 0$$

$$x(1-x) = 0$$

$$x=0 \quad x=1$$

$\notin$
Domain



$f''(x)$	$(-\infty, 0)$	$(0, 1)$	$(1, \infty)$
$\frac{2(3x^2 - 3x + 1)}{(x - x^2)^3}$	+	+	+
	-	+	-
	∩	∪	∩

$\therefore f(x)$ is concave upward on $(0, 1)$

$\therefore f(x)$ is concave downward on $(-\infty, 0)$ $(1, \infty)$

$\Rightarrow$ No inflection point

Key points

H.A

$$0 = \frac{1}{x-x^2}$$

* (x - x^2)

$$0 = 1$$

None

$$\begin{array}{c|cc|c} -1 & \frac{1}{2} & \frac{1}{4} & 2 \\ \hline 0 & & & 1 \end{array}$$

at $x = -1$

$$y = \frac{1}{-1-1} = \frac{1}{-2} = -\frac{1}{2}$$

$$\Rightarrow (-1, -\frac{1}{2})$$

at $x = \frac{1}{4}$

$$y = \frac{1}{\frac{1}{4} - \frac{1}{16}} = \frac{1}{\frac{4}{16} - \frac{1}{16}} = \frac{1}{\frac{3}{16}}$$

$$= \frac{16}{3}$$

$$\Rightarrow (\frac{1}{4}, \frac{16}{3})$$

at $x = 2$

$$y = \frac{1}{2-4} = \frac{1}{-2} = -\frac{1}{2}$$

$$\Rightarrow (2, -\frac{1}{2})$$

$$D_f = \mathbb{R} \setminus \{0, 1\}$$

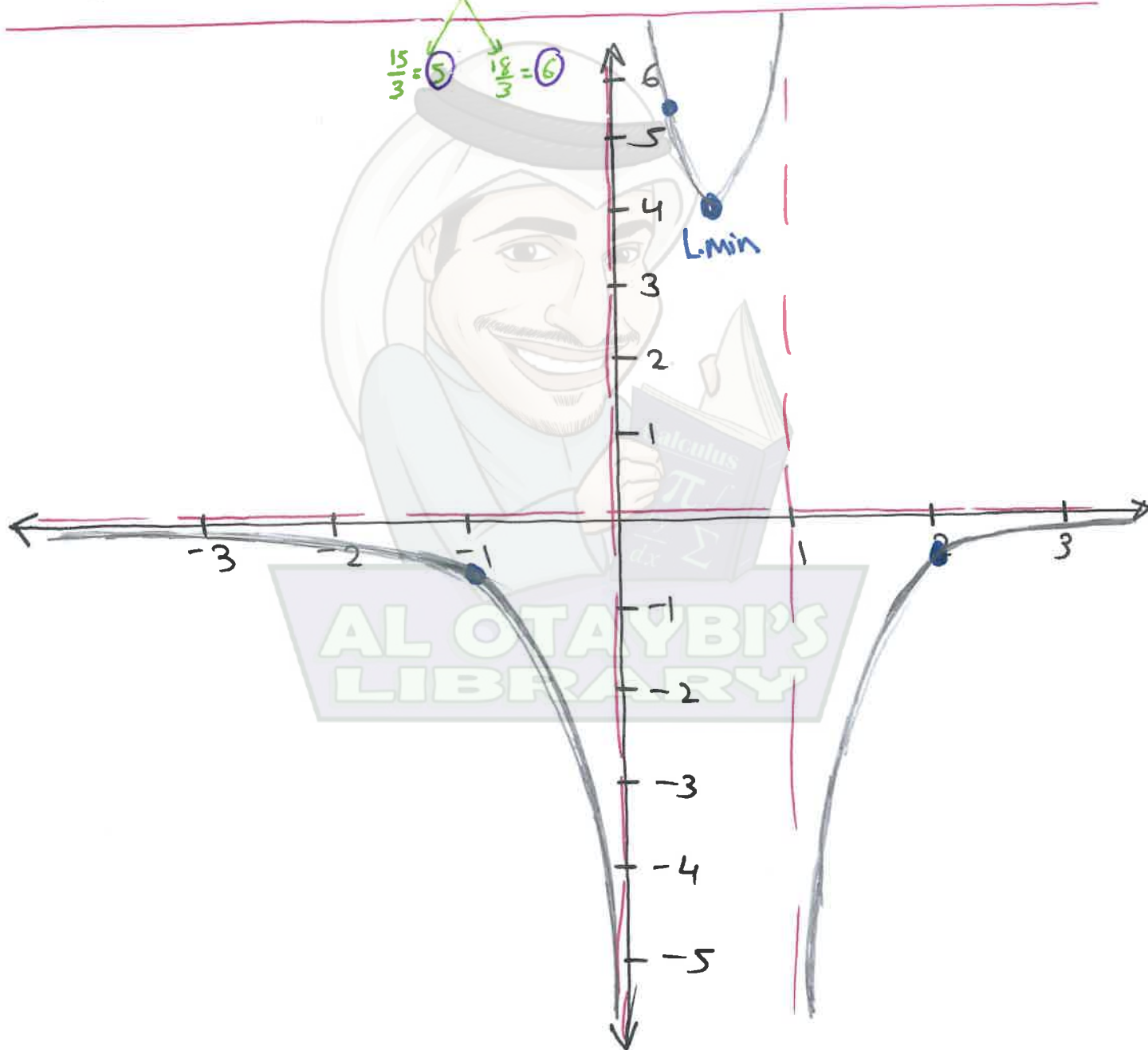
$$HA \rightarrow y=0$$

$$V.A \begin{cases} x=0 & \begin{cases} \oplus & +\infty \\ \ominus & -\infty \end{cases} \\ x=1 & \begin{cases} \oplus & -\infty \\ \ominus & +\infty \end{cases} \end{cases}$$

$$\left(\frac{1}{2}, 4\right) \text{ L.min}$$

$$\cup (0, 1) \quad \cap (-\infty, 0) \quad (1, \infty)$$

$$\left(-1, -\frac{1}{2}\right) \quad \left(\frac{1}{4}, \frac{16}{3}\right) \quad \left(2, -\frac{1}{2}\right) \quad \text{key points}$$



Let $f(x) = \frac{4 - x^2}{x^2 - 9}$

- (a) Study the symmetry of the curve and find the x-intercept and y-intercept, if any.
- (b) Find the vertical and horizontal asymptotes for the graph of f , if any.

Given that $f'(x) = \frac{10x}{(x^2 - 9)^2}$

- (c) Find the intervals on which f is increasing and the intervals in which f is decreasing.
- (d) Find the local maximum and local minimum values of f , if any.

Given that $f''(x) = \frac{-30(x^2 + 3)}{(x^2 - 9)^3}$

- (e) Find the intervals on which f concave upward and the intervals on which f is concave downward.
- (d) Find the points of inflection of f , if any.
- (f) Sketch the graph of f .

Past exam



هناك سؤال
مخشوش !!



$$f(x) = \frac{4-x^2}{x^2-9}$$

Domain of $f(x)$: $x^2 - 9 = 0$
 $(x-3)(x+3) = 0$
 $x = 3 \quad x = -3$

$$\Rightarrow \therefore D_f = \mathbb{R} \setminus \{\pm 3\}$$

H.A

$$\lim_{x \rightarrow \pm \infty} \frac{4-x^2}{x^2-9}$$

$$\lim_{x \rightarrow \pm \infty} \frac{\cancel{x^2}(\frac{4}{x^2}-1)}{\cancel{x^2}(1-\frac{9}{x^2})} = \frac{0-1}{1-0} = \frac{-1}{1} = -1$$

$$\Rightarrow \therefore y = -1 \text{ is H.A}$$

V.A

$$x = 3$$

$$\lim_{x \rightarrow 3^+} \frac{4-x^2}{(x-3)(x+3)} = \frac{4-9}{0^+} = \frac{-5}{0^+} = -\infty$$

$$\lim_{x \rightarrow 3^-} \frac{4-x^2}{(x-3)(x+3)} = \frac{4-9}{0^-} = \frac{-5}{0^-} = +\infty$$

$$\Rightarrow \therefore x = 3 \text{ is V.A}$$

$$x = -3$$

$$\lim_{x \rightarrow -3^+} \frac{4-x^2}{(x-3)(x+3)} = \frac{4-9}{0^-} = \frac{-5}{0^-} = +\infty$$

$$\lim_{x \rightarrow -3^-} \frac{4-x^2}{(x-3)(x+3)} = \frac{4-9}{0^+} = \frac{-5}{0^+} = -\infty$$

$$\Rightarrow \therefore x = -3 \text{ is V.A}$$

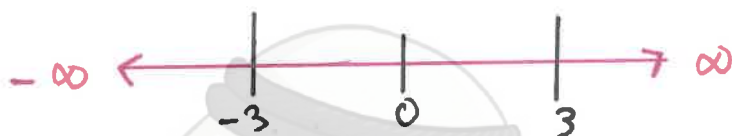
$$f'(x) = \frac{10x}{(x^2-9)^2}$$

= 0

$$\begin{aligned} 10x &= 0 \\ \div 10 \quad x &= 0 \in \text{Domain} \end{aligned}$$

DNE

$$\begin{aligned} x^2 - 9 &= 0 \\ (x-3)(x+3) &= 0 \\ x &= 3 \quad x = -3 \\ &\notin \text{Domain} \end{aligned}$$



$f'(x)$	$(-\infty, -3)$	$(-3, 0)$	$(0, 3)$	$(3, \infty)$
$\frac{10x}{(x^2-9)^2}$	-	-	+	+
	+	+	+	+



- $\therefore f(x)$ is increasing on $(0, 3)$ and $(3, \infty)$
- $\therefore f(x)$ is decreasing on $(-\infty, -3)$ and $(-3, 0)$

at $x=0 \rightarrow$ there is Local minimum

$$y = \frac{4-0}{0-9} = \frac{4}{-9} = -\frac{4}{9}$$

$\Rightarrow \therefore (0, -\frac{4}{9})$ Local minimum.

No Local maximum point

$$f''(x) = \frac{-30(x^2 + 3)}{(x^2 - 9)^3}$$

$= 0$

DNE

$$\begin{aligned} -30(x^2 + 3) &= 0 \\ \div (-30) & \\ x^2 + 3 &= 0 \\ x^2 &= -3 \\ \text{None} \end{aligned}$$

$$\begin{aligned} x^2 - 9 &= 0 \\ (x-3)(x+3) &= 0 \\ x &= 3 \quad x = -3 \\ &\notin \text{Domain} \end{aligned}$$



$f''(x)$	$(-\infty, -3)$	$(-3, 3)$	$(3, \infty)$
$\frac{-30(x^2+3)}{(x^2-9)^3}$	—	—	—
	+	—	+
	⌒	∪	⌒

- $\therefore f(x)$ is concave upward on $(-3, 3)$
 - $\therefore f(x)$ is concave downward on $(-\infty, -3)$ and $(3, \infty)$
- $\Rightarrow$ No inflection point.

Key points

HA

$$-1 = \frac{4-x^2}{x^2-9}$$

$$-(x^2-9) = 4-x^2$$

$$-x^2+9 = 4-x^2$$

$$-x^2+9+x^2-4=0$$

$$5=0$$

None.



at $x = -4$

$$y = \frac{4-16}{16-9} = -\frac{12}{7} \rightarrow (-4, -\frac{12}{7})$$

at $x = 1$

$$y = \frac{4-1}{1-9} = \frac{3}{-8} = -\frac{3}{8} \rightarrow (1, -\frac{3}{8})$$

at $x = 4$

$$y = \frac{4-16}{16-9} = -\frac{12}{7} \rightarrow (4, -\frac{12}{7})$$

* The symmetry *

$$f(-x) = \frac{4-(-x)^2}{(-x)^2-9}$$

$$= \frac{4-x^2}{x^2-9}$$

$$= f(x)$$

$$\therefore f(-x) = f(x)$$

$\therefore f(x)$ has symmetry about y-axis.

* Intercept *

X-intercept

$$0 = \frac{4-x^2}{x^2-9}$$

$$*(x^2-9)$$

$$0 = 4-x^2$$

$$x^2 = 4$$

$$x = 2, x = -2$$

$$(-2, 0) (2, 0)$$

Y-intercept

$$y = \frac{4-0}{0-9}$$

$$y = \frac{4}{-9} = -\frac{4}{9}$$

$$(0, -\frac{4}{9})$$

$$D_f = \mathbb{R} \setminus \{\pm 3\}$$

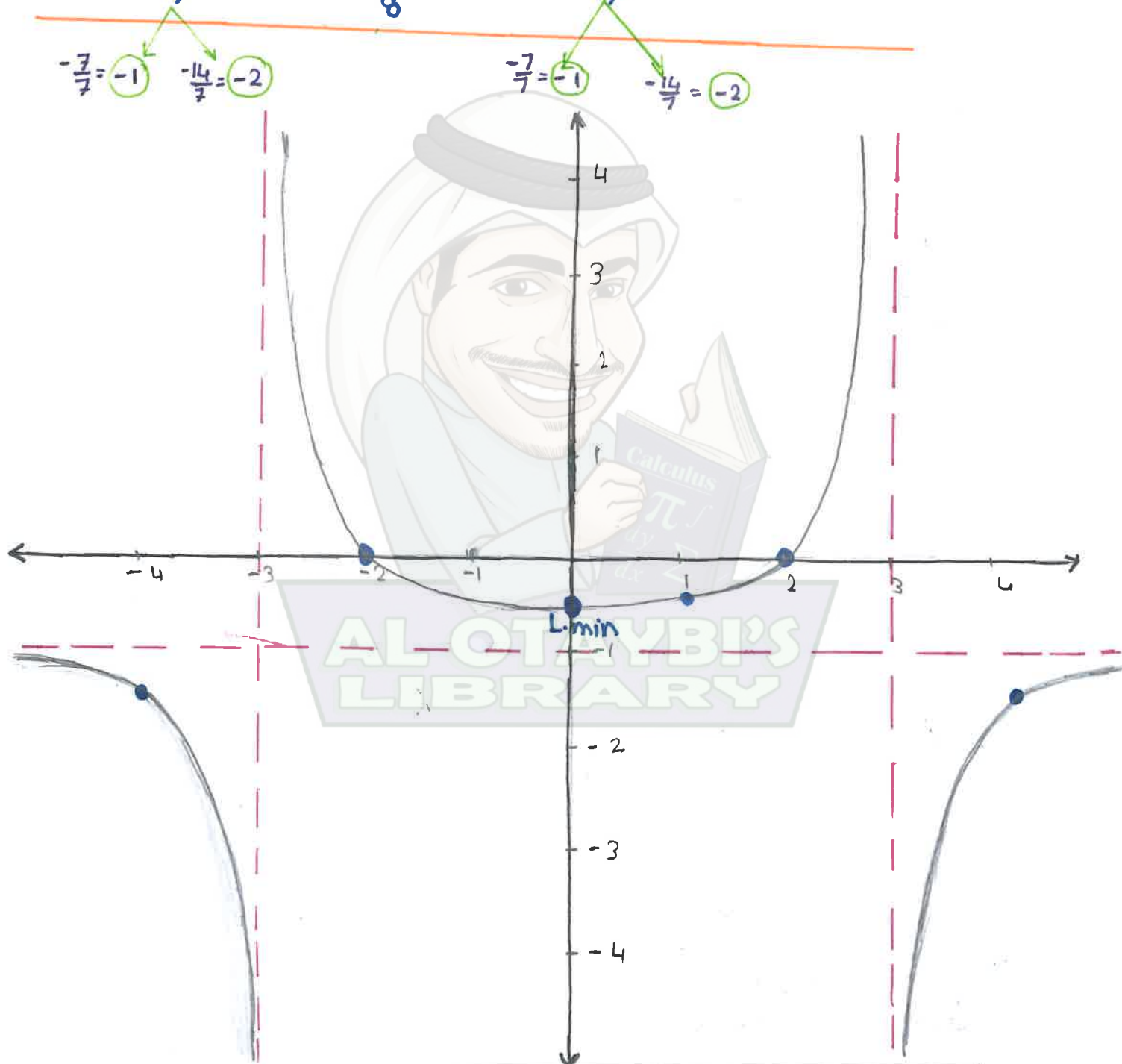
$$H.A \rightarrow y = -1$$

$$V.A \begin{cases} x = 3 \begin{cases} \oplus & -\infty \\ \ominus & +\infty \end{cases} \\ x = -3 \begin{cases} \oplus & +\infty \\ \ominus & -\infty \end{cases} \end{cases}$$

$$(0, -\frac{4}{9}) \text{ L.min}$$

$$\cup (-3, 3) \quad \cap (-\infty, -3) \quad (3, \infty)$$

$$(-4, -\frac{12}{7}) \quad (1, -\frac{3}{8}) \quad (4, -\frac{12}{7}) \quad \text{key points}$$





Tricky question !!



Given that $f(x) = \frac{x^2 - 3}{x + 2}$

(a) Find the vertical and horizontal asymptotes for the graph of f , if any.

Given that $f'(x) = \frac{x^2 + 4x + 3}{(x + 2)^2}$

(b) Find the intervals on which f is increasing and the intervals in which f is decreasing.

(c) Find the local maximum and local minimum values of f , if any.

Given that $f''(x) = \frac{2}{(x + 2)^3}$

(d) Find the intervals on which f concave upward and the intervals on which f is concave downward.

(e) Find the points of inflection of f , if any.

(f) Sketch the graph of f .

(g) Find the absolute extrema of $f(x)$ on $\left[-\frac{3}{2}, 0\right]$

Past exam



هناك سؤال

مخشوش !!



$$f(x) = \frac{x^2 - 3}{x + 2}$$

Domain of $f(x)$: $x + 2 = 0$
 $x = -2$

$$\Rightarrow \therefore D_f = \mathbb{R} \setminus \{-2\}$$

H.A

$$\lim_{x \rightarrow \pm\infty} \frac{x^2 - 3}{x + 2} = \lim_{x \rightarrow \pm\infty} \frac{x^2 [1 - \frac{3}{x^2}]}{x [1 + \frac{2}{x}]}$$

$$\lim_{x \rightarrow \pm\infty} \frac{x [1 - \frac{3}{x^2}]}{1 + \frac{2}{x}} = \frac{\pm\infty [1 - 0]}{1 + 0} = \pm\infty$$

$$\Rightarrow \therefore \text{No H.A}$$

V.A

$$\lim_{x \rightarrow -2^+} \frac{x^2 - 3}{x + 2} = \frac{4 - 3}{0^+} = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow -2^-} \frac{x^2 - 3}{x + 2} = \frac{4 - 3}{0^-} = \frac{1}{0^-} = -\infty$$

$$\Rightarrow \therefore x = -2 \text{ is V.A}$$

$$f'(x) = \frac{x^2 + 4x + 3}{(x+2)^2}$$

$= 0$

$$x^2 + 4x + 3 = 0$$

$$(x+1)(x+3) = 0$$

$$x = -1 \quad x = -3$$

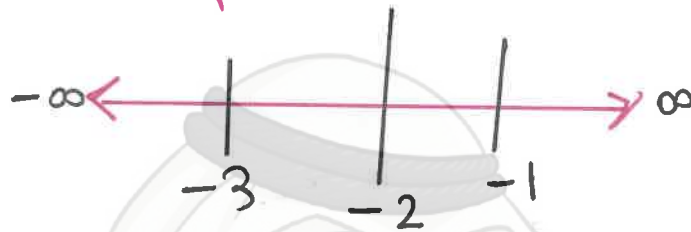
$\in \text{Domain}$

DNE

$$x+2=0$$

$$x = -2$$

$\notin \text{Domain}$



$f'(x)$	$(-\infty, -3)$	$(-3, -2)$	$(-2, -1)$	$(-1, \infty)$
$\frac{(x+1)(x+3)}{(x+2)^2}$	+	-	-	+
	+	+	+	+

(+)

(-)

(-)

(+)

$\therefore f(x)$ is increasing on $(-\infty, -3)$ and $(-1, \infty)$

$\therefore f(x)$ is decreasing on $(-3, -2)$ and $(-2, -1)$

at $x = -3 \rightarrow$ there is Local maximum

$$y = \frac{-3-3}{-3+2} = \frac{-6}{-1} = 6 \Rightarrow (-3, 6) \text{ L.max}$$

at $x = -1 \rightarrow$ there is Local minimum

$$y = \frac{-1-3}{-1+2} = \frac{-4}{1} = -4 \Rightarrow (-1, -4) \text{ L.min}$$

$$f''(x) = \frac{2}{(x+2)^3}$$

$$= 0$$

$$2 = 0$$

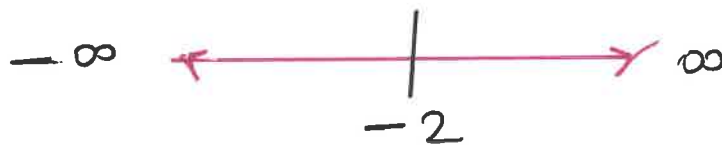
None

$$\text{DNE}$$

$$x+2=0$$

$$x=-2$$

$\notin \text{Domain}$



$f''(x)$	$(-\infty, -2)$	$(-2, \infty)$
$\frac{2}{(x+2)^3}$	+	+
	-	+
	$\ominus$	$\oplus$

$\therefore f(x)$ is concave upward on $(-2, \infty)$

$f(x)$ is concave downward on $(-\infty, -2)$

$\Rightarrow$ No inflection point.

key points

(HA)

None

$$-4 \quad x \quad | \quad 0$$

-2

at $x = -4 \longrightarrow y = \frac{-4-3}{-4+2} = \frac{-7}{-2} = \frac{7}{2}$

$\Rightarrow (-4, \frac{7}{2})$

at $x = 0 \longrightarrow y = \frac{0-3}{0+2} = -\frac{3}{2}$

$\Rightarrow (0, -\frac{3}{2})$

intersection

x-intercept

y-intercept

$$0 = \frac{x^2 - 3}{x + 2}$$

$\times (x + 2)$

$$0 = x^2 - 3$$
$$x^2 = 3$$

$$x = -\sqrt{3} \quad x = \sqrt{3}$$

$$(-\sqrt{3}, 0) \quad (\sqrt{3}, 0)$$

$$y = \frac{0-3}{0+2} = -\frac{3}{2}$$
$$(0, -\frac{3}{2})$$

$$D_f = \mathbb{R} \setminus \{-2\}$$

H.A $\rightarrow$ None

$$V.A \rightarrow x = -2 \begin{cases} \oplus & +\infty \\ \ominus & -\infty \end{cases}$$

$(-3, -6)$ L.max

$(-1, -2)$ L.min

$\cup (-2, \infty)$

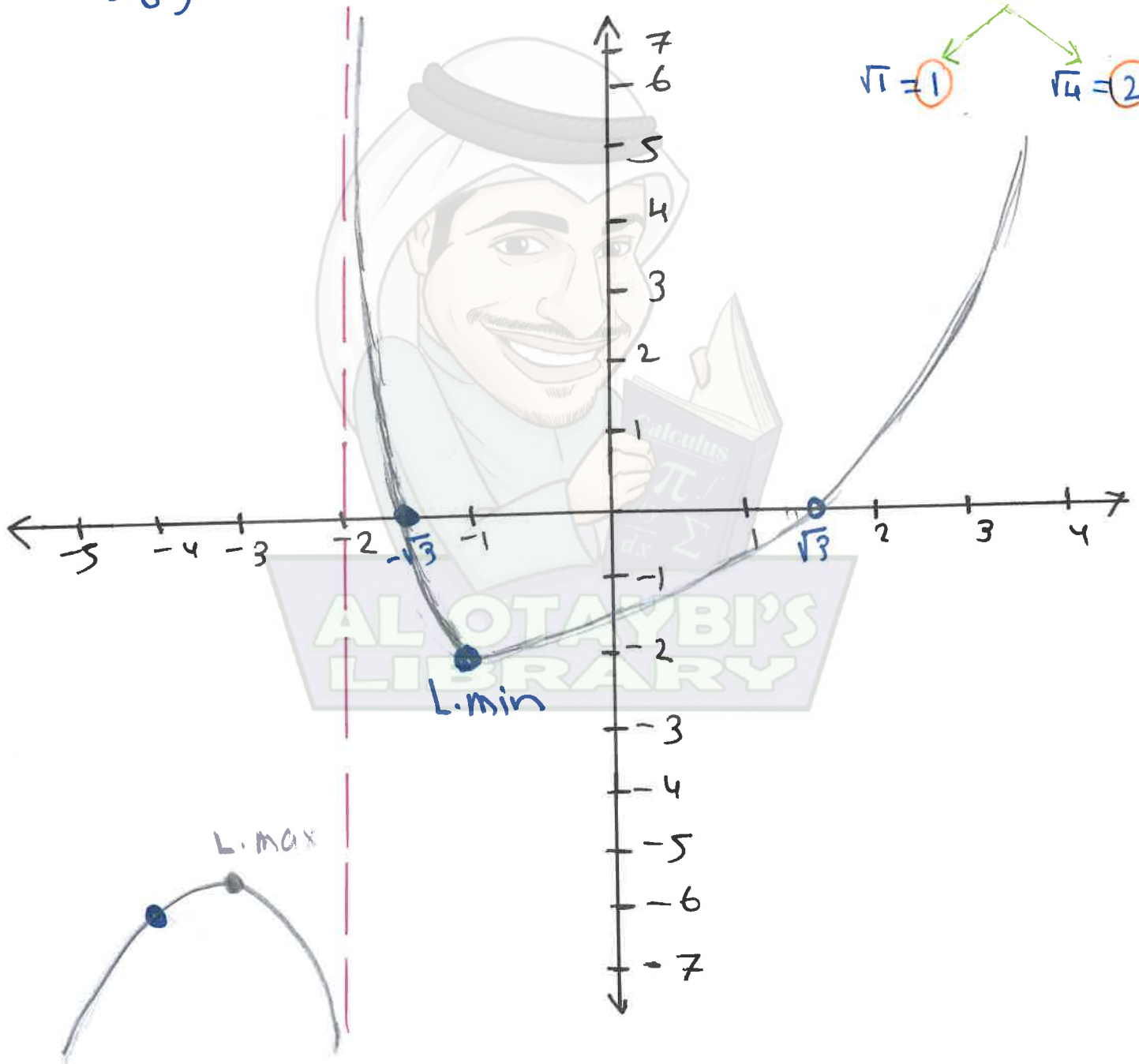
$\cap (-\infty, -2)$

$(-4, -\frac{13}{2})$ $(0, -\frac{3}{2})$

$\downarrow$
 -6.5

intercept
 $(\sqrt{3}, 0)$ $(-\sqrt{3}, 0)$
 $(0, -\frac{3}{2})$

$$\sqrt{3} \begin{cases} \sqrt{1} = 1 \\ \sqrt{4} = 2 \end{cases}$$



* سؤال منشور *

$$f'(x) = \frac{x^2 + 4x + 3}{(x+2)^2} \quad \left[-\frac{3}{2}, 0\right]$$

= 0

$$x^2 + 4x + 3 = 0$$

$$(x+1)(x+3) = 0$$

$$x = -1$$

$$x = -3$$

$$\in \left[-\frac{3}{2}, 0\right]$$

$$\notin \left[-\frac{3}{2}, 0\right]$$

DNE

$$x+2=0$$

$$x = -2$$

$$\notin \left[-\frac{3}{2}, 0\right]$$

$$\begin{aligned} f\left(-\frac{3}{2}\right) &= \frac{\left(-\frac{3}{2}\right)^2 - 3}{-\frac{3}{2} + 2} = \frac{\frac{9}{4} - 3}{-\frac{3}{2} + 2} \\ &= \frac{\frac{9}{4} - \frac{12}{4}}{-\frac{3}{2} + \frac{4}{2}} = \frac{-\frac{3}{4}}{\frac{1}{2}} = -\frac{3(2)}{4} = \left(-\frac{3}{2}\right) \end{aligned}$$

$$f(0) = \frac{0 - 3}{0 + 2} = \left(-\frac{3}{2}\right)$$

$$f(-1) = \frac{1 - 3}{-1 + 2} = \frac{-2}{1} = (-2)$$

∴ the absolute maximum is $-\frac{3}{2}$

the absolute minimum is -2

Consider the function $f(x) = \frac{e^x}{x^2}$

a) Find the intervals, if any, on which f is increasing or decreasing.

b) Find the local maxima and minima of f , if any.

Jan 2017

Domain of $f(x) = \mathbb{R} \setminus \{0\}$

Vertical Asymptote

$x=0$ is V.A

$$f'(x) = \frac{x^2 e^x - e^x (2x)}{(x^2)^2} = \frac{x e^x [x-2]}{x^4}$$

$$= \frac{e^x [x-2]}{x^3}$$

$= 0$

$$e^x [x-2] = 0$$

$$e^x = 0$$

None

$$x-2=0$$

$$x=2$$

$\in$ Domain

undefined

$$x^3 = 0$$

$$x=0$$

$\notin$ Domain



	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
$\frac{e^x [x-2]}{x^3}$	-	+	+
	-	+	+

$\oplus$

$\ominus$

$\oplus$

a) interval of increasing
 $= (-\infty, 0)$ and $(2, \infty)$

interval of decreasing
 $= (0, 2)$

b) at $x=2 \rightarrow$ Local minimum

$$y = \frac{e^2}{4}$$

$(2, \frac{e^2}{4})$ is Local minimum

No Local Maximum Point